(Please write your Enrollment number immediately) Enrollment No. ----------Hrs.

SUPPLEMENTARY MID TERM EXAMINATION

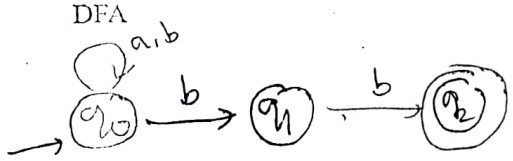
B.TECH PROGRAMMES (UNDER THE AEIS OF USICT)

IV Semester, March 2025

Paper Code: CIC-206 Theory of Computation

Time: $1\,{}^{1}/_{2}$ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No		Question	Max. Marks	CO(s)
1	(a)	Define regular expressions recursively	2	CO1
	(b)	Write regular expression for strings containing the pattern "*abac*" over $\sum = \{a, b, c\}$	2	CO1
	(c)	Write CFG for $L = \{a^n b^m \mid m > n\}$	2	CO2
	(d)	Define ambiguity in CFG. What is inherent ambiguity in CFL?	2	CO2
	(e)	Define Chomsky hierarchy of languages	2	CO1 & CO 2
2	(a)	Define DFA and NFA formally. Convert the following NFA to DFA 	5	CO1
	(b)	Show that $L = \{a^n b^n \mid n > 0\}$ is not regular using pumping lemma	5	CO1
3	(a)	Define a Pushdown Automaton. Write PDA for $L = \{wcw \mid w \in (a+b)^*\}$	5	CO2
	(b)	Define normal forms in CFG. Convert the following to CNF $S \rightarrow bA \mid aB, A \rightarrow bAA \mid aS \mid a, B \rightarrow aBB \mid bS \mid b$	5	CO2
4	(a)	Convert to GNF: $A_1 \rightarrow A_2A_3, A_2 \rightarrow A_3A_1 \mid b, A_3 \rightarrow A_1A_2 \mid a$	5	CO2
	(b)	Prove that the following language is not context free a. $L = \{a^{n^2} b^n\}$	5	CO2